

A Data Driven Approach to Retail Price Classification & Optimization using online APIs

By:

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Abstract

This study examines how online retailers set prices, use promotions, and adjust discounts, and how these behaviors influence demand. Using daily pricing data from two major retailers, the project classifies each retailer's pricing strategy through engineered features such as volatility, rigidity, and promotional activity. A demand strength index is developed to estimate relative demand in the absence of sales data. A Random Forest model then predicts demand responses under different pricing scenarios, forming a practical prescriptive system. The findings offer managers clearer insight into pricing behavior and data-driven guidance for optimizing promo and price decisions.

Keywords:

Retail pricing, demand estimation, prescriptive analytics, promotions, discount depth, machine learning, dynamic pricing, EDLP, Random Forest, price optimization.

Introduction

Pricing is one of the most important decisions retailers make. A small change in price can shift demand, change how customers view a product, and directly shape revenue and profit. Today, prices change quickly, promotions come and go, and customers compare options across multiple online platforms within seconds. Because of this, retailers can no longer rely only on fixed pricing rules or intuition. They need tools that help them understand their own pricing behavior, how it compares to competitors, and how customers might respond when prices or promotions change.

This project focuses on building such a system using real pricing data collected from online retailers. The study begins by examining how retailers behave—whether their prices stay stable, change often, or rely heavily on promotions. These patterns reflect well-known pricing strategies such as Everyday Low Pricing, Hi-Lo promotion strategies, or more dynamic approaches. By tracking day-to-day price movements, discount depth, and promotional activity, the study identifies the underlying strategy each retailer appears to use.

However, knowing a retailer's pricing strategy is not enough. Managers also need to understand how these actions affect customer interest. Because direct sales data were not available, the study creates a demand strength index using signals from pricing patterns, promotions, and competitive positioning. This index acts as a stand-in for demand and allows the study to model how different pricing decisions might influence customer response.

Finally, the project brings these pieces together in a prescriptive system. Using machine learning, the system predicts how demand might change at different price or promotion levels. This allows retailers to test scenarios—such as increasing discounts or adjusting prices relative to MSRP—and see which choices are likely to strengthen demand.

Overall, the goal of this project is to provide a clear, data-driven way for retailers to understand and improve their pricing decisions.

Literature Review

2.1 Overview and Purpose of the Review

This section reviews past research related to retail pricing, data-driven analytics, prescriptive analytical systems, and explainable artificial intelligence. The goal is to understand how retailers make pricing decisions, how data analytics has evolved to support these decisions, and how explainability and human collaboration are becoming essential parts of analytical systems. The review builds the foundation for this study by identifying where current research falls short and how this project contributes to filling that gap.

2.2 Evolution of Retail Pricing Strategies

Retail pricing has been studied because it directly affects how consumers perceive value and how companies earn profits. The four common strategies:

- **Everyday Low Pricing (EDLP)** – stable and consistent prices designed to build trust and reduce uncertainty.
- **Hi-Lo Pricing** – frequent promotions and discounts to attract customers temporarily.
- **Dynamic Pricing** – flexible prices that change frequently in response to demand or competition.
- **Commitment Pricing** – pre-announced markdown paths or long-term pricing commitments.

Studies such as Park, Jung and Choi (2020) and Gerpott et al. (2022) explain how these strategies are chosen based on market conditions, competition, and consumer response. Dynamic pricing has become a central strategy in modern retail because demand, competition, and inventory conditions fluctuate continuously, making fixed pricing inefficient in many markets. Under dynamic pricing, retailers frequently adjust prices allowing them to capture higher margins during peak demand and clear inventory through timely markdowns when demand weakens (Park et al., 2020). In contrast, pricing-commitment behaviors such as EDLP or long-term price guarantees reflect a retailer's strategic choice to maintain stable, prices. Together, dynamic pricing and pricing-commitment strategies illustrate two distinct competitive logics shape consumer expectations and reduce search costs.

Retailers often adopt an Everyday Low Pricing (EDLP) strategy because its competitive strength stems from operational efficiency. Park, Jung, and Choi (2020) show that the equilibrium price for EDLP retailers depends primarily on cost structures meaning that EDLP stores compete by maintaining lean operations instead of reacting to weekly promotional swings. Therefore, we can surmise that EDLP pricing behavior depends mainly on operational costs rather than competitor behavior.

A Hi-Lo pricing strategy is adopted by a retailer because consumer demand in many retail categories is highly responsive to temporary discounts and promotional activity. Park, Jung, and Choi (2020) show that the equilibrium price for Hi-Lo retailers is strongly shaped by their promotional decisions, particularly the size of the promotional basket—the proportion of items offered at a discount. A wider basket substantially increases Hi-Lo store sales while simultaneously reducing sales at nearby EDLP competitors, making it a powerful competitive lever (Park et al., 2020).

Although these models describe pricing behaviors, most are descriptive—they explain what retailers are doing but not what they should do to improve results. Schlosser and Boissier (2018) argue that even large online retailers face difficulties adjusting prices optimally under competition because traditional models cannot handle complex, real-time data.

These findings show that while pricing strategy theory is well developed, research still lacks systems that help managers make prescriptive, data-driven decisions that adapt automatically to market conditions.

2.3 Data-Driven Retail Pricing Strategy Classification

Classifying retailers by pricing strategy is essential because different strategies respond differently to price and promotional adjustments. Therefore, the classification literature provides the foundation for building the prescriptive analytical system used in this study. The extant literature shows that a rule-based classification system based on historical pricing data using engineered features is the most reliable method

Price volatility is widely recognized in the pricing literature as a key behavioral indicator of dynamic pricing strategies. Retailers that frequently modify prices in response to demand fluctuations, competitive pressure, or inventory changes exhibit characteristically high volatility.

Elmaghraby and Keskinocak (2003) describe dynamic retailers as those who continually update prices to reflect market conditions. Similarly, Fassnacht and Unterhuber (2016) identify high-frequency price adjustments as a defining feature of dynamic-pricing orientation, distinguishing these retailers from EDLP operators who maintain stable price paths. Chintagunta et al. (2009) further argue that volatility serves as a measurable signal of competitive intensity and retailer responsiveness. More recently, Park, Jung, and Choi (2020) show that online retailers competing in fast-moving digital environments often rely on short-term, reactive pricing adjustments, reinforcing the role of volatility as a valid classifier of pricing strategy. Collectively, this literature supports the use of volatility-based metrics such as rolling price variation, change frequency, and price-range dispersion as theoretically grounded indicators for distinguishing EDLP retailers from dynamic or Hi-Lo retailers.

Promotional activity is another central dimension through which retailer pricing strategies can be distinguished, particularly when contrasting EDLP and Hi-Lo formats. Early work by Hoch, Dreze, and Purk (1994) demonstrated that Hi-Lo retailers rely heavily on deep, frequent discounts to generate store traffic. Whereas EDLP retailers minimize promotional activity and instead emphasize stable everyday prices. Lal and Rao (1997) reinforced this distinction by showing that EDLP retailers typically maintain shallow discount depths and low promotion frequency. On the other hand, Hi-Lo retailers systematically employ large markdowns and broader promotional breadth. Ailawadi, Neslin, and Gedenk (2001) further found that promotion intensity and discount depth are strongly aligned with retailers' strategic objectives, with Hi-Lo retailers allocating significantly more budget and shelf space to promotional events. Narasimhan, Neslin, and Sen (1996) observed that promotion variability itself is a structural signature of Hi-Lo strategies, reflecting a deliberate reliance on episodic discounting. More recent work by Park, Jung, and Choi (2020) empirically supports the use of promo breadth, promo depth, and promotional volatility as behavioral signals that identify retailer competitive positioning. Overall, this literature establishes that promotion-based metrics are theoretically grounded indicators of retailer pricing philosophy, making variables such as promotion intensity, discount depth, and promotional volatility appropriate for classifying retailers into EDLP, Hybrid, or Hi-Lo strategies.

2.4 Data-Driven Retail Price Optimization

With the growth of e-commerce and big data, retailers began using machine learning (ML) to forecast demand and guide pricing. Models such as linear regression, random forest, and gradient boosting help predict how price changes affect demand and revenue. In that context it is important to understand which pricing strategy would be best for a certain retailer therefore, the literature review continues by exploring retail price optimization.

Retail price optimization is increasingly important in modern retail markets because product demand is highly sensitive to many factors. Such as price levels, the dynamic interactions between competing products, time-varying consumer preferences, and market conditions. The paper by Subbarayudu et al., 2023 argues that traditional manager-driven pricing is no longer sufficient. As retailers must process large volumes of historical sales data to uncover subtle, nonlinear demand patterns that cannot be identified through intuition alone. The authors emphasize that accurate pricing is critical because price directly affects demand, competitive position, and overall profitability. Machine learning provides a systematic way to forecast demand and compute optimal prices by capturing complex relationships between price movements and customer behavior. Among tested models, the study finds that random forest regression performs best (Subbarayudu et al., 2023). As it achieves higher R^2 , lower MSE, and lower maximum error than multiple linear regression (Subbarayudu et al., 2023). Similarly, Schlosser and Boissier (2018) used ML-based demand estimation to develop automated pricing strategies that outperform manual rule-based systems.

This shift led to what Ito and Fujimaki (2016) call the “predict-then-optimize” approach, where data scientists first build a predictive model of demand and then use optimization techniques to recommend the best pricing decision. However, most of these systems remain focused on prediction accuracy rather than decision quality or transparency.

Retail demand is often unobservable at the SKU level in competitive online environments, leading researchers to rely on proxy-based demand estimation when sales data are unavailable. empirical pricing research indicates that competitive price gaps (e.g., price vs. MSRP) and promotional intensity correlate with shifts in consumer responsiveness and product attractiveness (Ailawadi et al., 2001; Lal & Rao, 1997). Building on this literature, a composite demand index such as the DSI can meaningfully approximate SKU-level demand strength by integrating these

behavioral drivers into a unified metric, allowing for consistent comparison across retailers and products in the absence of direct sales data.

The main limitation in this stream of work is that data-driven optimization often operates as a black box. Managers may not understand how or why the system recommends a certain price, which limits trust and adoption in real organizations.

2.5 Prescriptive Analytics Systems (PAS)

Prescriptive analytics represents the next stage in analytics maturity after descriptive and predictive analytics. According to Lepenioti et al. (2020), prescriptive systems aim not only to predict outcomes but also to recommend concrete actions that achieve desired goals. Wissuchek et al. (2024) extend this definition by viewing PAS as socio-technical systems that combine computational power with human judgment.

Ito and Fujimaki (2016) provide an early example of prescriptive price optimization, where algorithms suggest optimal prices using predicted demand curves. Daase (2025) adds that automation is transforming pricing decisions from manual tasks into continuous optimization cycles guided by analytics.

Yet, these systems face practical challenges. Many organizations lack transparency in how recommendations are generated, and human decision-makers may resist acting on opaque system outputs. Thus, even though PAS frameworks exist, their success depends on explainability and effective human collaboration—areas that remain underexplored, especially in retail settings.

2.6 Explainable AI and Human–AI Collaboration

Explainable Artificial Intelligence (XAI) has emerged as an important response to the black-box problem in machine learning. Linardatos et al. (2020) and Hassija et al. (2024) describe XAI as a set of methods that help users understand how AI models make predictions, such as by showing which variables most influence an outcome. These techniques—like feature importance or SHAP values—allow analysts to interpret results rather than blindly trust model outputs.

Bauer et al. (2023) studied how explainability affects human sense-making and found that transparent AI systems increase trust and acceptance. Meske and Bunde (2022) further highlight that organizations adopting AI must consider different stakeholder needs for transparency and accountability.

When applied to prescriptive systems, explainability helps transform analytics from a purely technical function into a collaborative decision process. Wissuchek et al. (2024) emphasize that effective human–AI collaboration happens when the system provides understandable reasoning and humans provide contextual judgment.

In retail analytics, an explainable PAS can show not only what price is optimal but also why—for example, by linking recommendations to changes in demand, discount depth, or volatility. This builds trust and helps managers make informed, accountable decisions.

Data & Methods

This project uses three methodological components:

- (1) a rule-based classification of retailer pricing strategy,
- (2) construction of a Demand Strength Index (DSI), and
- (3) development of an elasticity-based demand model using Random Forest regression.

Together, these steps form an end-to-end prescriptive analytical system (PAS) that can diagnose a retailer's pricing approach and recommend optimized promotional and pricing decisions.

3.1 Rule-Based Classification of Retailer Pricing Strategy

3.1.1 Motivation and Theoretical Basis

Retailers typically follow one of two dominant pricing strategies:

- **Everyday Low Pricing (EDLP)** – stable prices, minimal discount depth, very low-price volatility.
- **High–Low (Hi-Lo) Pricing** – frequent price changes, temporary deep promotions, wider discount ranges.

This distinction is widely described in pricing and retail analytics literature. EDLP retailers aim for stable expectations and operational efficiency, while Hi-Lo retailers use promotions to generate short-term spikes in traffic, compete dynamically, or clear inventory.

Since my dataset contains daily price histories for individual SKUs from Walmart and eBay, it allows me to measure how prices evolve over time. These price trajectories directly reflect each retailer's pricing philosophy.

This is why a rule-based classification approach is appropriate: instead of relying on assumptions, I let the retailer’s actual patterns of price behavior determine the classification.

3.1.2 Data Collection and Preparation

Daily headphone and earbuds product data was collected through:

- Walmart Affiliate Marketing API
- eBay Browse API.

This was done by gathering SKUs from the Walmart API and then matching as many of those SKUs to the eBay Browse API. Allowing for 173 completely identical SKU data to be collected for analysis. The figure below shows some of the SKUs gathered in a tabular comparison form.

UPC	Retailer	Product Title	Price (USD)	Capture Date
190199801561	ebay	Beats - Flex Wireless Earphor	49.99	2025-11-25
190199801561	walmart		30.01	2025-11-25
194252484296	ebay	Beats Fit Pro - True Wireless t	175.95	2025-11-25
194252484296	walmart		149.0	2025-11-25
194644039165	ebay	Soundcore K20i Semi-in-Ear t	29.99	2025-11-25
194644039165	walmart		19.99	2025-11-25
194644126155	ebay	Soundcore - by Anker P25i Tr	24.99	2025-11-25
194644126155	walmart		19.99	2025-11-25
194644186333	ebay	Soundcore P30i Noise Cancel	29.99	2025-11-25
194644186333	walmart		24.99	2025-11-25

Figure 1

The data includes SKU-level snapshots of:

- current price
- regular price
- discount depth
- shipping cost
- stock flags
- MSRP
- timestamp (“captured_at”)

	captured_at	retailer_id	native_item_id	upc	title_raw	brand_raw	model_raw	category_raw_path	product_url	currency	...
0	2025-10-02 22:45:38+00:00	ebay	v1 112358941238 412775517839	046838068768	JVC Gummy Headphones New in Retail Sealed Pack...	JVC	HAF-160	<NA	https://www.ebay.com/itm/112358941238?var=4127...	USD	...
1	2025-10-02 22:45:38+00:00	ebay	v1 116390435229 0	0812887017213	JLab - GO Air POP True Wireless In- Ear Headpho...	JLab	EBGAIRPOPRLT124	<NA	https://www.ebay.com/itm/116390435229?amdata=e...	USD	...
2	2025-10-02 22:45:38+00:00	ebay	v1 116746209631 0	050036395403	JBL Tune Beam - True Wireless Active Noise Can...	JBL	None	<NA	https://www.ebay.com/itm/116746209631	USD	...
3	2025-10-02 22:45:38+00:00	ebay	v1 135564241444 0	<NA>	JLAB JBuds Pro Signature Earbuds w/Universal M...	JLab	EPRORBLU123, 6211313	<NA	https://www.ebay.com/itm/135564241444?amdata=e...	USD	...
4	2025-10-02 22:45:38+00:00	ebay	v1 145917048227 0	810045688787	Skullcandy S2TAW- Q740 Smokin Buds XT True Wir...	Skullcandy	S2TAW-Q740	<NA	https://www.ebay.com/itm/145917048227	USD	...

5 rows x 44 columns

Figure 2. Raw Data.

I cleaned and standardized this data and engineered several price-evolution features used in the classification. The data cleaning involved:

- Convert timestamps to datetime
- Sort data chronologically
- Remove duplicate rows
- Filter unrealistic price values
- Engineer new features (price_change, volatility, discount_depth, etc.)
- Aggregate to SKU-level when needed
- Handle missing values (as per needed)

3.1.3 Feature Engineering for Classification

To classify pricing strategy, I computed features that I determined through the literature review to distinguish Hi-Lo vs. EDLP behavior.

These include:

1. Price Change Frequency

How often prices change for each SKU.

$$\text{change_freq} = \text{mean}(\text{price_change} \neq 0)$$

2. Rolling 30-Day Price Volatility

$$\text{roll_vol_30} = \text{std}(\text{price_current over 30 days})$$

3. Discount Range

$$\text{discount_range} = \max(\text{discount_depth}) - \min(\text{discount_depth})$$

4. Promo Volatility

Day-to-day variability in overall promotional activity.

5. Number of Unique Prices per SKU

Dynamic retailers typically exhibit larger price distributions.

6. EDLP Consistency Metric

Measures stability between regular and current prices:

$$\text{edlp_consistency} = 1 - \frac{|\text{price_regular} - \text{price_current}|}{\text{price_regular}}$$

7. Rigidity / No-Change Rate

Proportion of days in which a SKU's price does not change.

retailer_id	native_item_id	change_freq	roll_vol_30	discount_range	promo_volatility	unique_prices	edlp_consistency	rigidity	price_range_metric
ebay	v1 112358941238 412775517839	0.000000	0.000000	0.000000	0.000000	1	NaN	0.981818	0.000000
ebay	v1 116390435229 0	0.000000	0.000000	0.000000	0.000000	1	NaN	0.981818	0.000000
ebay	v1 116746209631 0	0.000000	0.000000	0.000000	0.000000	1	NaN	0.981818	0.000000
ebay	v1 135564241444 0	-0.033333	0.127503	0.200222	0.428764	2	0.799778	0.709091	0.239944
ebay	v1 145917048227 0	0.000000	0.000000	0.000000	0.000000	1	NaN	0.981818	0.000000
...	...	...	...	...	...	...	...	...	...
walmart	8922602673	-0.185185	0.070902	0.333444	0.373355	2	0.673650	0.927273	0.364961
walmart	938592087	0.000000	0.055916	0.178027	0.373355	5	0.975392	0.800000	0.137572
walmart	962793502	0.000000	0.000000	0.000000	0.000000	1	0.666444	0.981818	0.000000
walmart	976661885	-0.000370	0.032646	0.100050	0.000000	7	0.508090	0.872727	0.128513
walmart	988529517	-0.183148	0.001695	0.131080	0.000000	3	0.392643	0.945455	0.011969

Figure 3. Engineered Features.

To begin the process of classification these features were first computed at the SKU level and then later aggregated at the retailer level to classify retailers.

3.1.4 Rule Construction and Application

Using these engineered features, I applied a transparent rule-based decision logic:

- If dynamic behavior metrics (change frequency, volatility, discount breadth) are high, retailer is labeled Hi-Lo.
- If stability metrics (EDLP consistency, rigidity) are high, retailer is labeled EDLP.
- If both dimensions fall in the middle, retailer is classified as Hybrid.

The classification was applied by computing the median aggregated scores for each retailer:

```
def classify(row, margin=0.3):  
    if row["dynamic_score"] > row["edlp_score"] + margin:  
        return "Dynamic"  
    if row["edlp_score"] > row["dynamic_score"] + margin:  
        return "EDLP"  
    return "Hybrid"  
  
retailer_scores["strategy"] = retailer_scores.apply(classify, axis=1)
```

Figure 4. The Classification formula.

3.2 Construction of the Demand Strength Index (DSI)

3.2.1 Motivation

In many retail datasets—especially online data scraped or retrieved through public APIs—actual sales volumes are not available. The literature shows that most demand prediction models require a demand proxy and sales volumes or inventory data such as stockout flags on SKUs act as the demand proxies. However, when pulling data from APIs, retailers do not provide sales volumes, and they do not mark in-stock flags. This creates a limit within the development of my prescriptive system. Thus, limiting the direct estimation of classic demand curves.

To address this, literature proposes using proxy signals of consumer demand, such as:

- Promotion intensity
- Discount depth
- Price competitiveness
- Rigidity

- Price volatility

These variables reflect how the market (and the retailer) respond to demand patterns.

I therefore built a Demand Strength Index (DSI)—a composite score that synthesizes multiple demand-related signals into a single interpretable variable.

This DSI serves two purposes:

- It becomes a target variable for elasticity modeling.
- It provides a retailer-level snapshot of demand strength across SKUs.

3.2.2 Components of the DSI

The DSI is constructed using six SKU-level behavioral metrics:

1. ***Promo Intensity***

Fraction of days a SKU is on promotion.

Higher = more effort to stimulate demand.

2. ***Average Discount Depth***

How deep discounts tend to be.

Reflects demand responsiveness.

3. ***Price Volatility***

High volatility typically indicates retailers adjusting to demand or competitive pressure.

4. ***Mean Price Competitiveness***

Price relative to MSRP (proxy for perceived value).

5. ***Rigidity (Price Stability)***

High rigidity = stable demand or EDLP behavior.

6. ***EDLP Consistency***

Captures how closely current prices track regular prices.

Each component was normalized using z-scores to place them on comparable scales. Because some metrics represent weak demand when high (e.g., discount_depth), I inverted them:

$\text{promo_intensity_inv} = -\text{promo_intensity}$

$\text{discount_depth_inv} = -\text{avg_discount_depth}$

After inversion, higher = better demand-conditions for all features.

retailer_id	native_item_id	promo_intensity	avg_discount_depth	price_vol	mean_price_vs_msrp	rigidity	edlp_consistency	demand_strength
walmart	233758219	0.000000	0.000000	0.000000	1.254024	0.981818	0.745976	1.179335
	482493465	0.000000	0.000000	0.000000	1.053823	0.981818	0.946177	1.080430
	416099849	0.000000	0.000000	0.003783	1.059575	0.963636	0.940425	1.063390
	252575372	0.000000	0.000000	0.000000	1.005030	0.981818	0.994970	1.056325
	55509211	0.000000	0.000000	0.000000	1.000000	0.981818	1.000000	1.053840
	479338409	0.000000	0.000000	0.000000	0.992662	0.981818	0.992662	1.044284
	13895355965	0.018182	0.012999	0.010977	1.002597	0.945455	0.971406	0.980220
	248390097	0.018182	0.005872	0.094288	1.101222	0.945455	0.887034	0.931136
	3752378948	0.090909	0.010866	0.014082	0.940238	0.927273	0.940238	0.852364
	1546280223	0.109091	0.015859	0.045947	0.984141	0.909091	0.984141	0.838759

Figure 5. A part of the normalized table showing Walmart SKUs

3.2.3 Composite DSI Calculation

The Demand Strength Index is defined as the average of normalized behavior signals:

$$DSI_i = \frac{z(\text{promo_intensity}) + z(\text{avg_discount_depth}) + z(\text{price_vol}) + z(\text{mean_price_vs_msrp}) + z(\text{rigidity}) + z(\text{edlp_consistency})}{6}$$

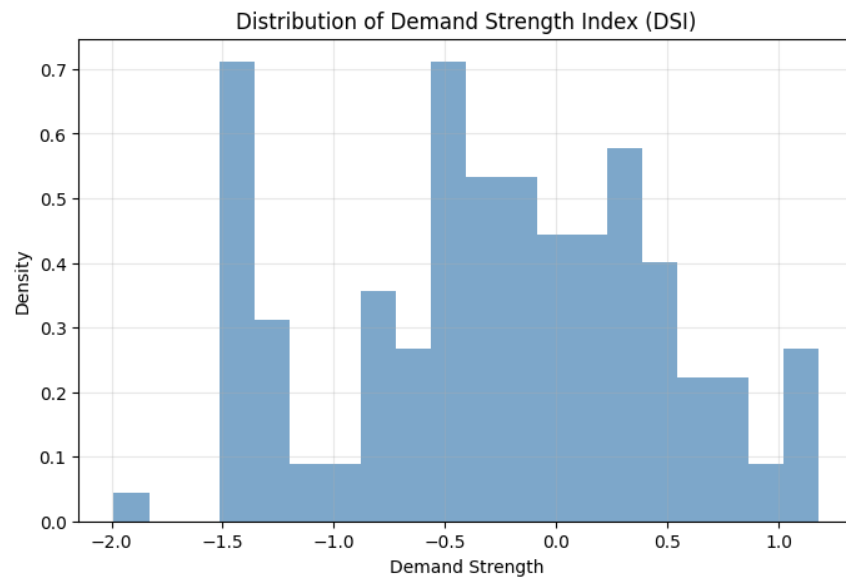


Figure 6. DSI is not uniform across SKUs. Meaning some SKUs have more demand compared to others.

This yields a continuous target variable representing relative demand strength across SKUs. And the resulting DSI positions each SKU on a continuous demand-strength spectrum:

- **High DSI** = High willingness-to-pay / strong latent demand
- **Low DSI** = Price-sensitive / promotion-dependent / weak demand

3.3 Elasticity-Based Demand Modeling (Random Forest Regression)

3.3.1 Motivation

Once the DSI is constructed, the next step is to model how price, discount depth, and promotional behavior influence demand strength.

Unlike simple linear elasticity formulas, a Random Forest Regression (RF) can capture:

- Nonlinear price effects
- Interactions between promotional variables
- Diminishing returns on deeper discounts
- Differences between EDLP and Hi-Lo retailers
- Complex SKU and retailer behavior patterns

This aligns with modern literature that shows tree-based methods perform well in retail price-demand forecasting when actual sales are unavailable.

Therefore, the RF model serves as the “engine” of my prescriptive analytical system.

3.3.2 Model Setup

Target variable

$$y = DSI$$

Feature set:

X

= {mean_price, mean_price_vs_msrp, price_vol, price_range_metric, promo_intensity,
avg_discount_depth, rigidity, edlp_consistency}

retailer_id	native_item_id	mean_price	mean_price_vs_msrp	price_vol	price_range_metric	promo_intensity	avg_discount_depth	rigidity	edlp_consistency	demand_strength
ebay	v1j112358941238412775517839	12.990000	0.713856	0.000000	0.000000	0.000000	0.000000	0.981818	0.748715	-0.234756
	v1j116390435229j0	12.950000	0.713856	0.000000	0.000000	0.000000	0.000000	0.981818	0.748715	-0.234756
	v1j116746209631j0	108.400000	0.713856	0.000000	0.000000	0.000000	0.000000	0.981818	0.748715	-0.234756
	v1j1135564241444j0	7.615455	0.713856	0.127503	0.239944	0.763636	0.152897	0.709091	0.799778	-0.234756
	v1j145917048227j0	20.000000	0.713856	0.000000	0.000000	0.000000	0.000000	0.981818	0.748715	-0.234756
...	...	...	...	...	...	...	...	...	...	...
walmart	8922602673	20.899091	0.673650	0.070902	0.364961	0.836364	0.278881	0.927273	0.673650	-0.198267
	938592087	48.652545	0.975392	0.055916	0.137572	0.163636	0.023238	0.800000	0.975392	0.687809
	962793502	9.990000	0.666444	0.000000	0.000000	1.000000	0.333556	0.981818	0.666444	-0.205176
	976661885	10.156727	0.508090	0.032646	0.128513	1.000000	0.491910	0.872727	0.508090	-0.661992
	988529517	29.624909	0.392643	0.001695	0.011969	1.000000	0.607357	0.945455	0.392643	-0.798378

Figure 7

The data was cleaned, aggregated to SKU level, and missing values were imputed using medians. A standard train–test split (80/20) was used.

3.3.3 Model Training and Performance

The Random Forest model achieved:

$R^2 \approx 0.96$ on the test set

```
from sklearn.metrics import r2_score

pred = rf.predict(X_test)
print("R²:", r2_score(y_test, pred))

✓ 0.0s
R²: 0.9634787795669073
```

Figure 8

This indicates the model captures most of the variance in DSI and can reliably estimate how demand responds to different pricing or promotional levels.

Feature Importance

The RF produced intuitive feature importances:

Feature	Importance
mean_price_vs_msrp	highest
avg_discount_depth	moderate
edlp_consistency	meaningful
promo_intensity	moderate
rigidity	small
price_vol	small
mean_price	small

Table 1

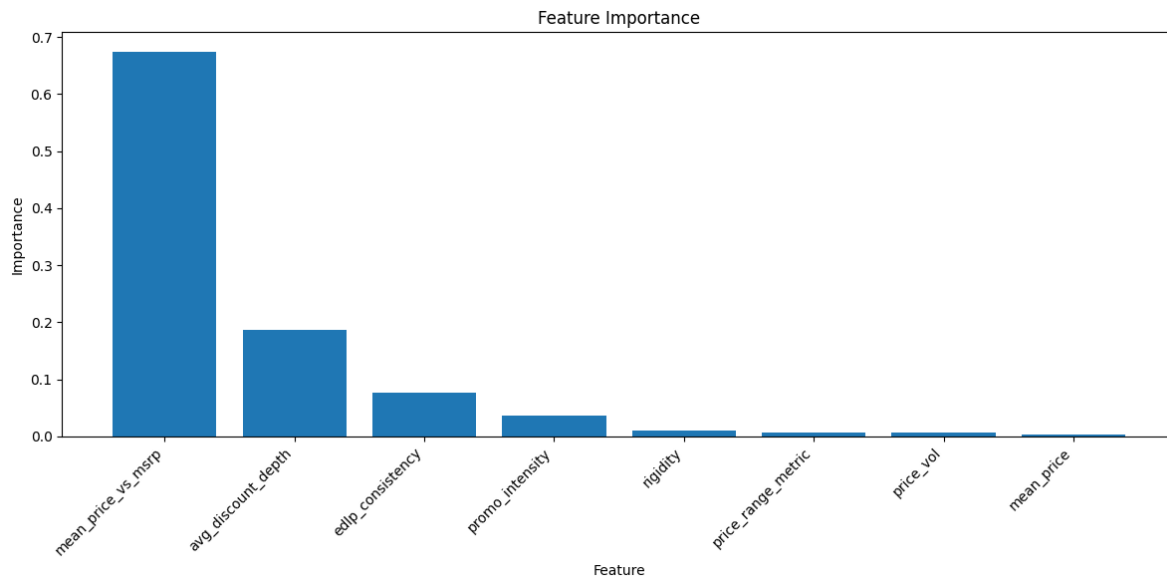


Figure 9

These results align with theory:

perceived value (price vs MSRP) and discount depth strongly influence demand.

3.3.4 Simulation and Prescription

Finally, I used the Random Forest model to run “what-if” price and promotion simulations, such as:

- **“What happens to demand_strength if we increase discount depth by 5%?”**

- **“What is the predicted change in demand if a retailer shifts toward a more EDLP-consistent pattern?”**

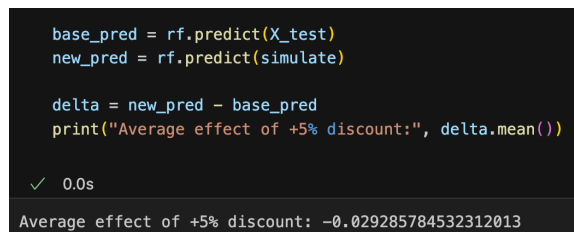
For example:

```
simulate = X_test.copy()

simulate["avg_discount_depth"] += 0.05

new_pred = rf.predict(simulate)

delta = new_pred - rf.predict(X_test)
```



```
base_pred = rf.predict(X_test)
new_pred = rf.predict(simulate)

delta = new_pred - base_pred
print("Average effect of +5% discount:", delta.mean())
```

✓ 0.0s

Average effect of +5% discount: -0.029285784532312013

Figure 10. The implications will be discussed in the next section.

This allows the model to act as a prescriptive engine, by estimating:

- Price elasticity
- Promo elasticity
- Optimal discount depth
- Optimal price competitiveness
- SKU-specific and retailer-level sensitivity

These simulations form the core of the Pricing Advisory System (PAS).

Results and Discussion

This chapter presents the empirical results of the study. It follows the structure of the methods:

- (1) exploratory data analysis (EDA),
- (2) rule-based retailer classification, and
- (3) elasticity-based demand modeling supported by the Demand Strength Index (DSI).

Together, these components offer a detailed view of how retailers behave, how their pricing patterns can be classified, and how demand can be inferred and simulated even without sales data.

4.1 Exploratory Data Analysis (EDA)

Daily price data was collected for a set of headphone and earbud SKUs across Walmart and eBay. Although both retailers carry overlapping UPCs, their pricing behavior differs sharply.

4.1.1 Walmart Pricing Behavior

The data shows that Walmart behaves like a structured, stable, EDLP-leaning retailer with periods of strong indication of Dynamic pricing behaviors especially through their promotions.

Key Observations

Highly stable prices but when volatility occurs it is high: The distribution of Walmart's price volatility is highly concentrated near zero, indicating that most SKUs experience minimal day-to-day price changes. This behavior is consistent with EDLP-style pricing, where retailers prioritize stability and predictability over frequent price fluctuations. Only a very small number of SKUs exhibit moderate volatility, suggesting occasional tactical adjustments hinting at periods of dynamic pricing.

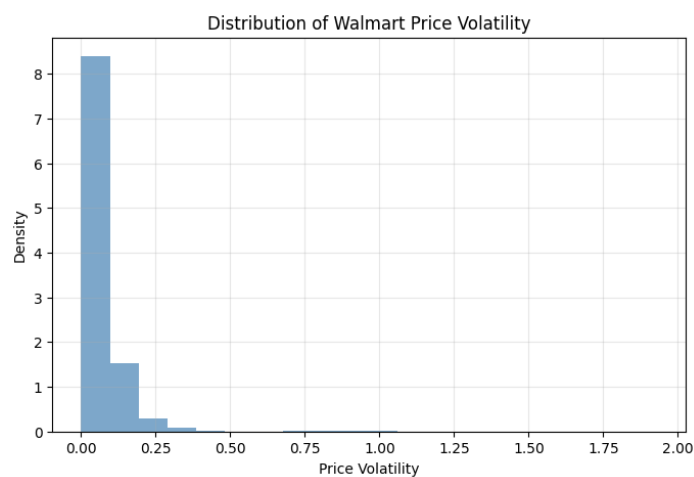


Figure 11. Prices barely over time and volatility is low

EDLP Consistency is Widely Distributed: The distribution of Walmart's EDLP Consistency scores shows a clear right-skewed pattern, with most values concentrated between 0.60 and 1.00.

This indicates that most Walmart's SKUs maintain prices that are very close to their reference (regular) price. In other words, Walmart rarely deviates from its baseline price, which is a defining feature of Everyday Low Pricing (EDLP) strategies.

However, the left side of the distribution from 0.10 to 0.50 reveals a non-trivial minority of SKUs that display moderate to high price variation. These could represent promotional cycles, clearance items, seasonal fluctuations, or SKUs where Walmart responds more dynamically to competitor pricing and is also known for its promotions. Thus, showing a tendency of Hi-Lo or Dynamic pricing behaviors.

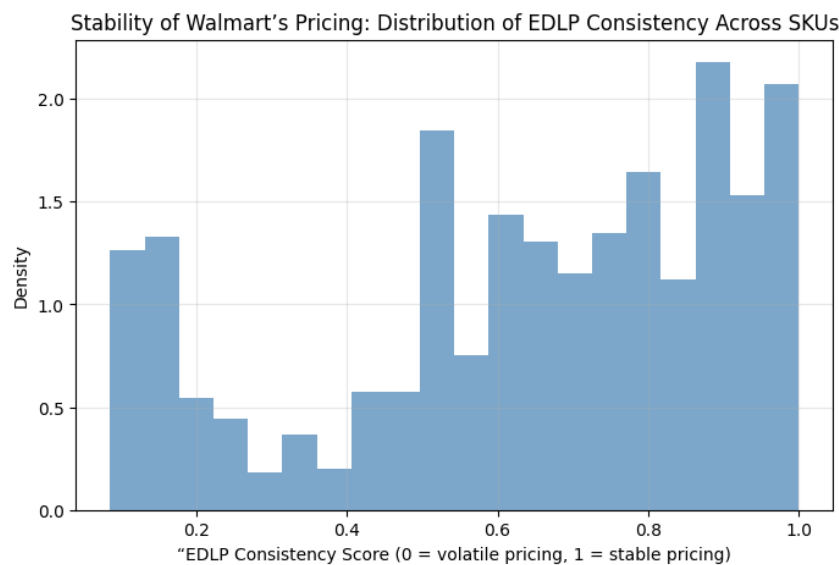


Figure 12

Moderate but structured promotions: This chart shows how five Walmart SKUs change in price over time. The pattern is clear: most prices stay flat for long periods, with only occasional adjustments. This is exactly what we expect from an EDLP-style retailer. A few SKUs show temporary dips, often short-lived and tied to known retail periods (e.g., early November promotions). These drops look like selective markdowns, not continuous dynamic pricing. Higher-priced items show bigger promotional swings, while cheaper items barely move at all. This suggests Walmart keeps strong rigidity in low-priced SKUs and only discounts higher-margin products occasionally. This shows that there are moderate but structured promotions and a tendency to show both EDLP and dynamic pricing behaviors.

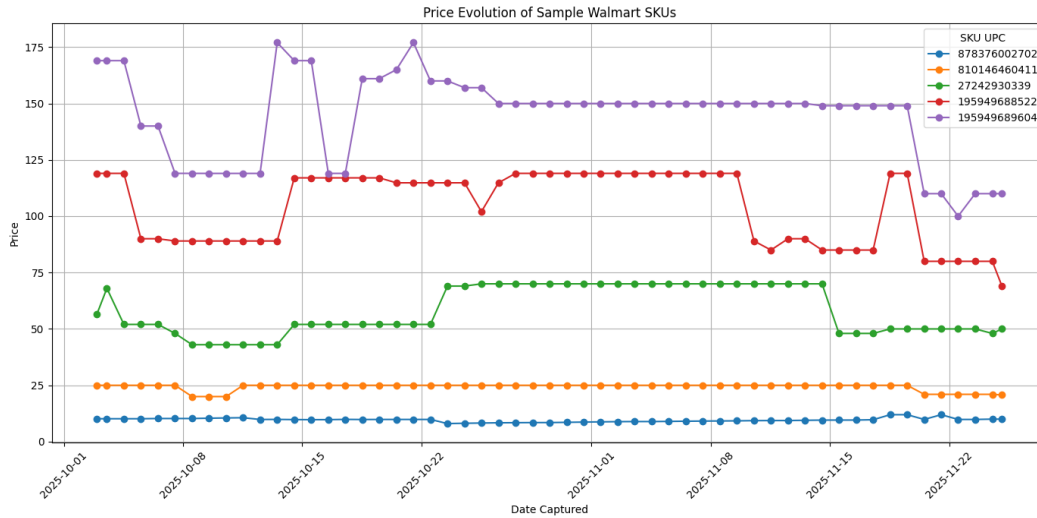


Figure 13

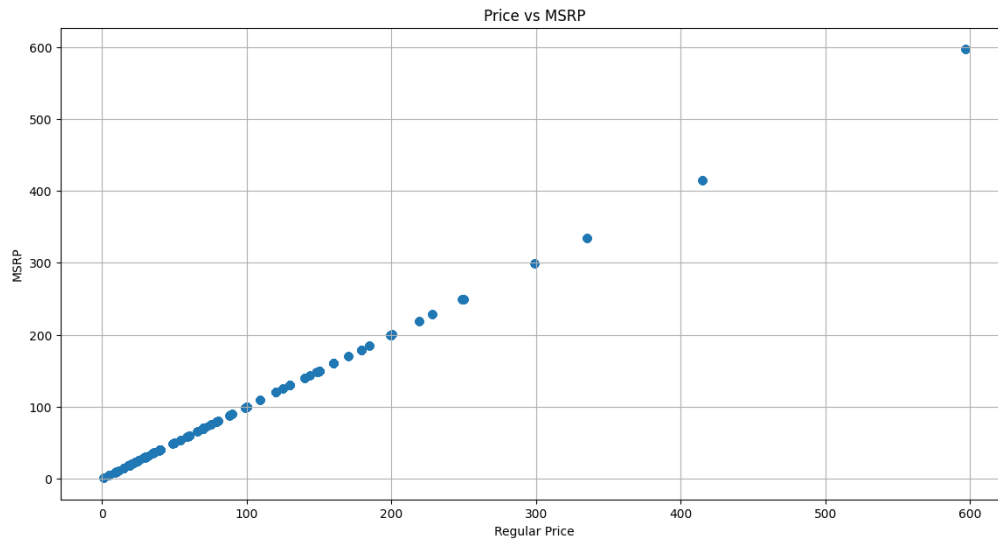


Figure 14

Tight anchor around MSRP: The Price vs MSRP scatter plot shows a near-perfect linear relationship, indicating that Walmart's pricing closely follows MSRP for most SKUs. Low-priced items cluster tightly around the line, suggesting minimal markdowns or variance. Higher-priced SKUs show slightly wider dispersion, reflecting occasional promotional adjustments or greater pricing flexibility in premium categories.

These visuals highlight how Walmart maintains consistent, predictable pricing with occasional controlled promotions.

4.1.2 eBay Pricing Behavior

eBay displays a near perfect description of an EDLP pricing strategy however, due to the existence of third-party sellers on a marketplace platform there are anomalies.

Key Observations

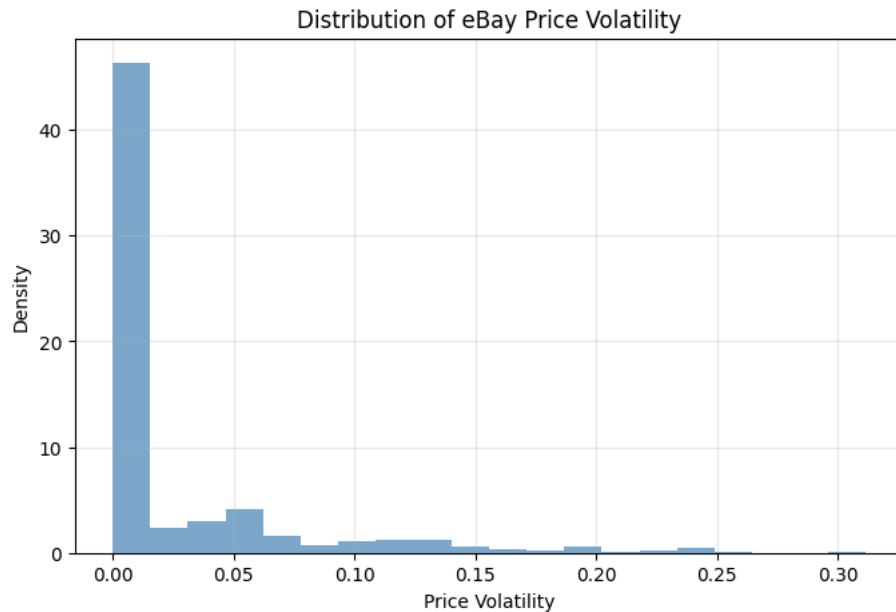


Figure 15

Consistent prices and low volatility with anomalies: The distribution of price volatility for eBay SKUs is heavily right-skewed. Most products exhibit very low day-to-day price variability (volatility below 0.02), indicating stable pricing for most listings. However, the long right tail reveals a subset of SKUs with substantially higher volatility — in some cases above 0.10 or even 0.30.

These highly volatile items likely reflect repricing behavior from third-party sellers, competitive pressure, or dynamic price adjustments common on marketplace-style platforms. This pattern contrasts with Walmart, where volatility is concentrated near zero with almost no long tail, however, the price shift is smaller compared to Walmart showing lower promotions or pricing changes.



Figure 16

Sparse promotional data: The line chart also shows the daily price evolution of five randomly selected eBay SKUs across the two-month collection period. All lines are almost perfectly flat — there are no promotions, no temporary discounts, and no noticeable price swings for any of the SKUs. This stability indicates extremely low-price volatility. Every SKU maintains the exact same price for the entire duration of the dataset.

In other words:

- 1) Prices do not respond to competitive pressure.
- 2) Prices do not change during seasonal events (including pre-Black-Friday).
- 3) Prices do not participate in Hi-Lo or dynamic promotional cycles.

This is exactly what an EDLP strategy looks like, even though eBay is traditionally a dynamic marketplace. In the category, sellers appear to favor fixed pricing, leading to behavior consistent with EDLP.

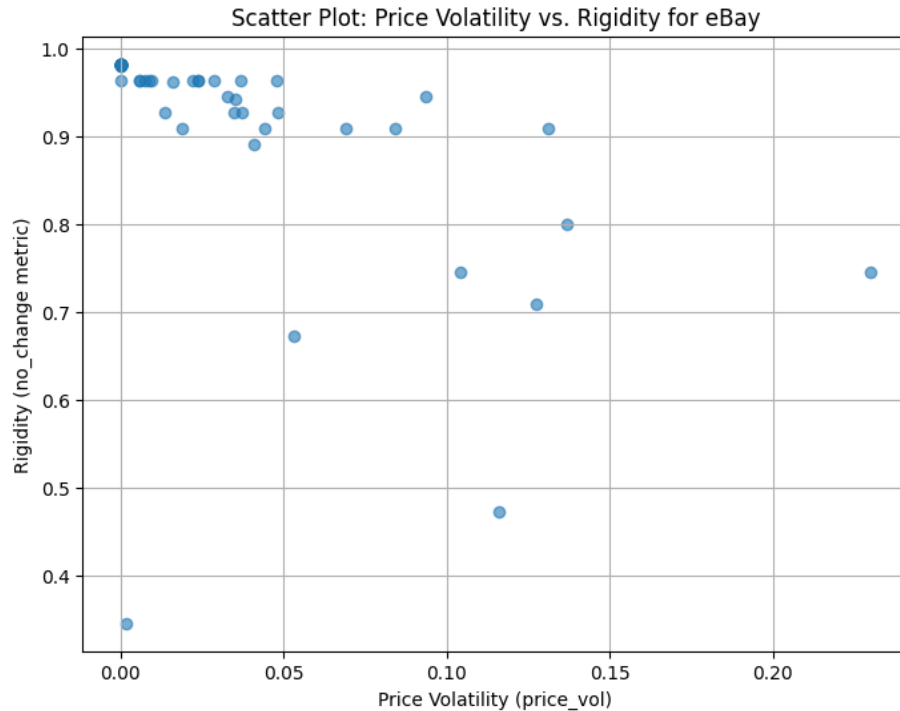


Figure 17. Most lie in the high rigidity/low-volatility region, indication highly stable prices with very few adjustments

Consistent rigidity: The scatter plot shows that almost all eBay SKUs cluster in the region of very low-price volatility (0.00–0.05) and very high rigidity (0.90–1.00). This means eBay prices rarely change and remain stable for long periods. Only a few SKUs show moderate volatility, and these appear to be isolated seller-level adjustments rather than a platform-wide pricing pattern.

Overall, the graph provides strong evidence that eBay follows an EDLP-like pricing behavior, characterized by stable, infrequently adjusted prices and minimal promotional activity. This visual pattern aligns with the rule-based classification results and reinforces the interpretation that eBay operates with a low-touch, steady pricing approach rather than a dynamic or Hi-Lo strategy.

4.1.3 Comparison Between Walmart and eBay

This subsection contrasts the two retailers using aggregated patterns.

Summary of Differences

Behavioral Trait	Walmart (Structured Retailer)	eBay (Marketplace)
Price Stability	Varying	High
Volatility	Low-Medium	Low
Promo Activity	Moderate & structured	None
EDLP Consistency	Varying	High
Price vs MSRP	Anchored	Not Captured
Rigidity	Mixed	Strong

Table 2

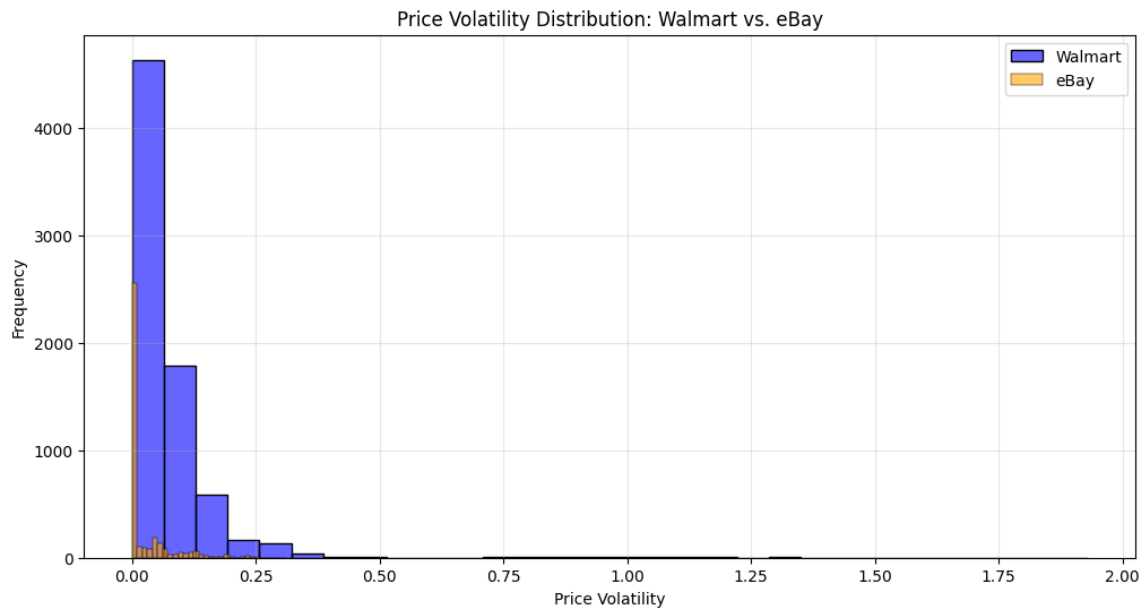


Figure 18. In a side-by-side comparison Walmart has a higher price volatility compared to eBay.

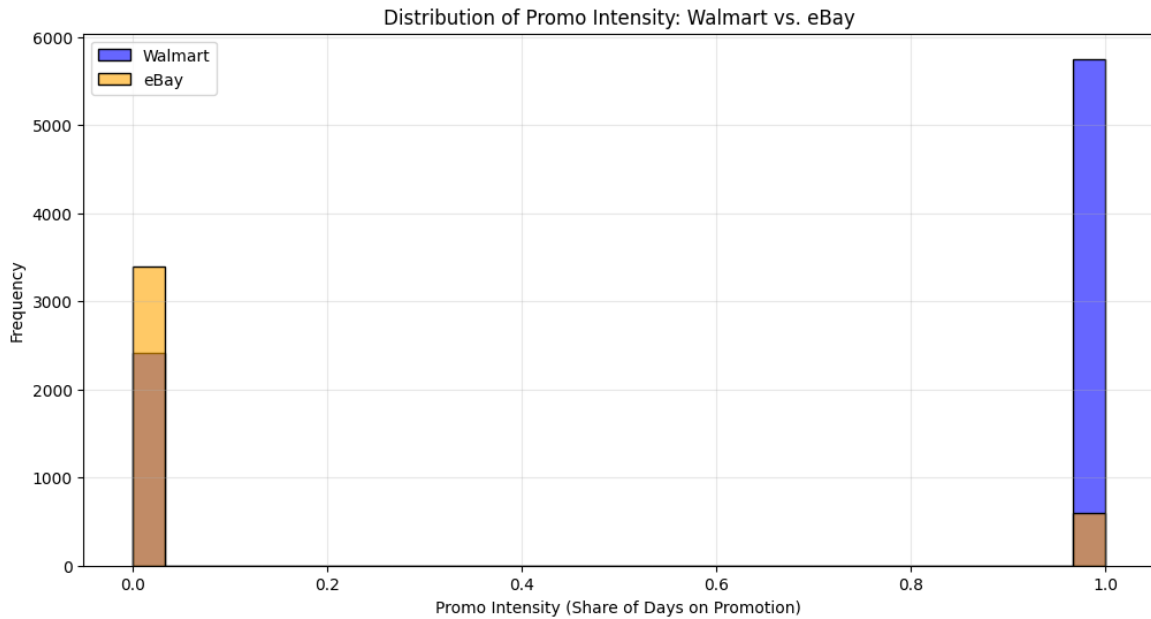


Figure 19. Walmart display far greater promotional consistency, with many SKUs on promotion nearly every day, whereas eBay shows very low or zero promo activity across its assortment.

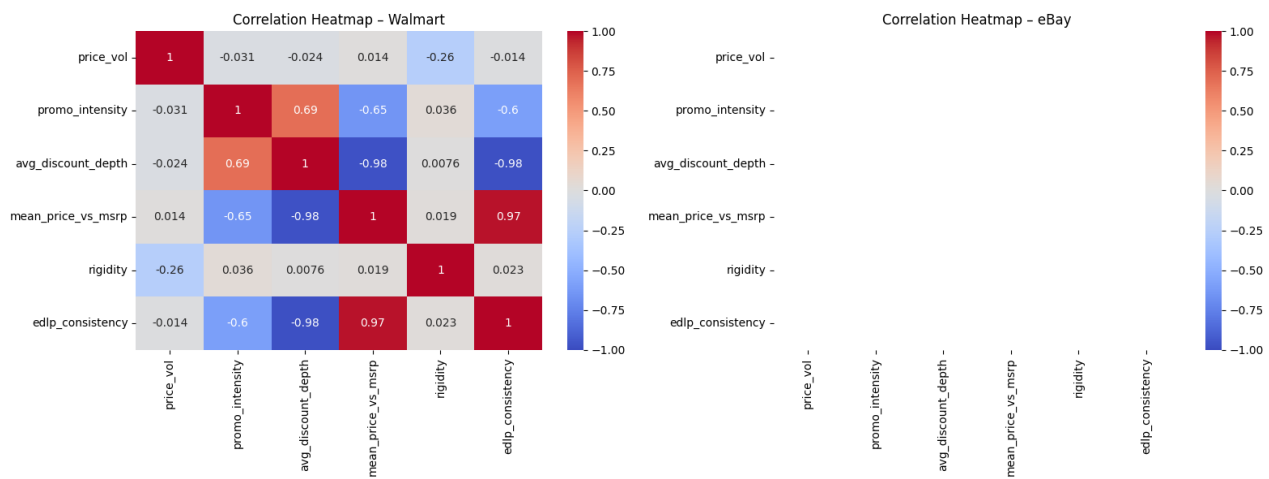


Figure 20

The Walmart heatmap reveals clear and meaningful correlation structure, which aligns with the behavior of a retailer that centrally coordinates pricing decisions:

Key Walmart Insights

- Promo Intensity and Discount Depth ($r \approx +0.69$):
SKUs that spend more days on promotion also receive deeper discounts.
This is a hallmark of a coordinated Hi-Lo component within a hybrid strategy.

- Discount Depth and EDLP Consistency ($r \approx -0.98$):
Deep discounts sharply reduce EDLP consistency, as expected — SKUs with large markdowns deviate from stable everyday pricing.
- Mean Price vs MSRP and EDLP Consistency ($r \approx +0.97$):
SKUs priced close to MSRP appear more EDLP-consistent.
This suggests Walmart maintains list-price stability for certain categories.
- Price Volatility and Rigidity ($r \approx -0.26$):
Items with lower volatility show higher rigidity, meaning their prices rarely change.
This again reflects a hybrid strategy: some SKUs are stable, others more dynamic.

Walmart's pricing variables move together in predictable ways. This strong internal correlation pattern is consistent with a centrally managed pricing model with elements of both EDLP and Dynamic adjustments — validating your rule-based classification of Walmart as Hybrid.

Key eBay Insights

eBay pricing data exhibits extremely low variance across most SKUs, with nearly constant prices and almost no recorded promotional activity. Since correlation requires variation in the underlying variables, the eBay dataset does not produce a stable correlation matrix. This reflects the nature of eBay's marketplace model: sellers set prices independently, promotions are infrequent or absent, and many SKUs do not experience enough price movement to compute volatility metrics. As a result, eBay shows no coherent multivariate pricing structure, supporting its rule-based classification as EDLP-leaning through price rigidity, but not due to corporate strategy — instead due to seller inertia.

These comparisons help frame why the classification results in later sections emerge the way they do.

4.2 Rule-Based Classification Findings

The rule-based classification system used engineered features to assign each retailer to one of three pricing strategies: Dynamic, EDLP, or Hybrid. The rules were grounded in pricing literature and applied directly to aggregated SKU-level behavior.

4.2.1 Classification Results

The system produced the following outcomes:

eBay classified as EDLP

Walmart classified as Hybrid

retailer_id	dynamic_score	edlp_score	strategy
ebay	-0.527378	0.554166	EDLP
walmart	0.005862	-0.097184	Hybrid

Figure 21

Why eBay classified as EDLP

Although eBay is a seller-driven marketplace, the rule-based system captures observable behavior rather than institutional strategy. eBay shows:

- Near-zero promo activity
- High rigidity for many SKUs
- Minimal discounting
- Sparse promotional signals
- Prices that often remain unchanged for long stretches

Under the defined rules, lack of promo behavior pushes eBay toward the EDLP category.

Why Walmart classified as Hybrid

Walmart shows features of both stability and occasional dynamic adjustments:

- Stable EDLP-like structure
- Small but regular price adjustments
- Structured promotions
- Moderate discount depth
- Slight volatility

This combination places Walmart in the Hybrid category.

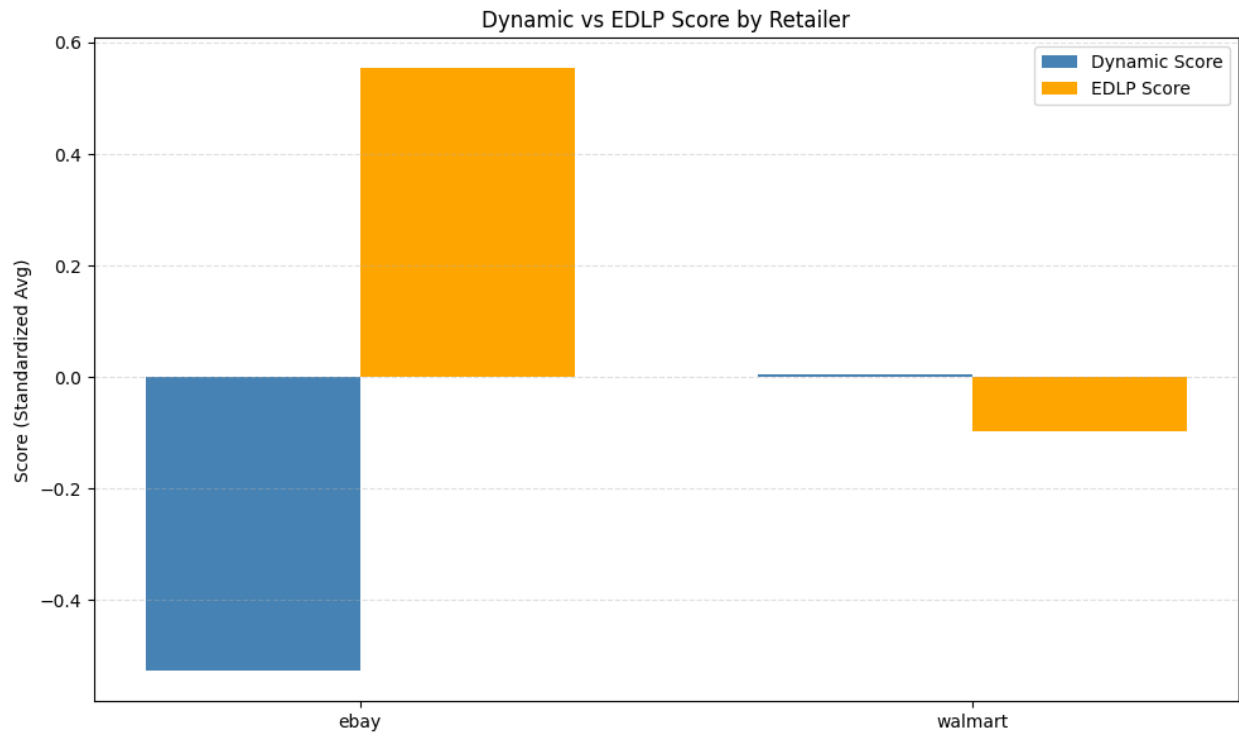


Figure 22. eBay has low dynamic score and high EDLP score whereas, Walmart has similar score for both becoming a Hybrid.

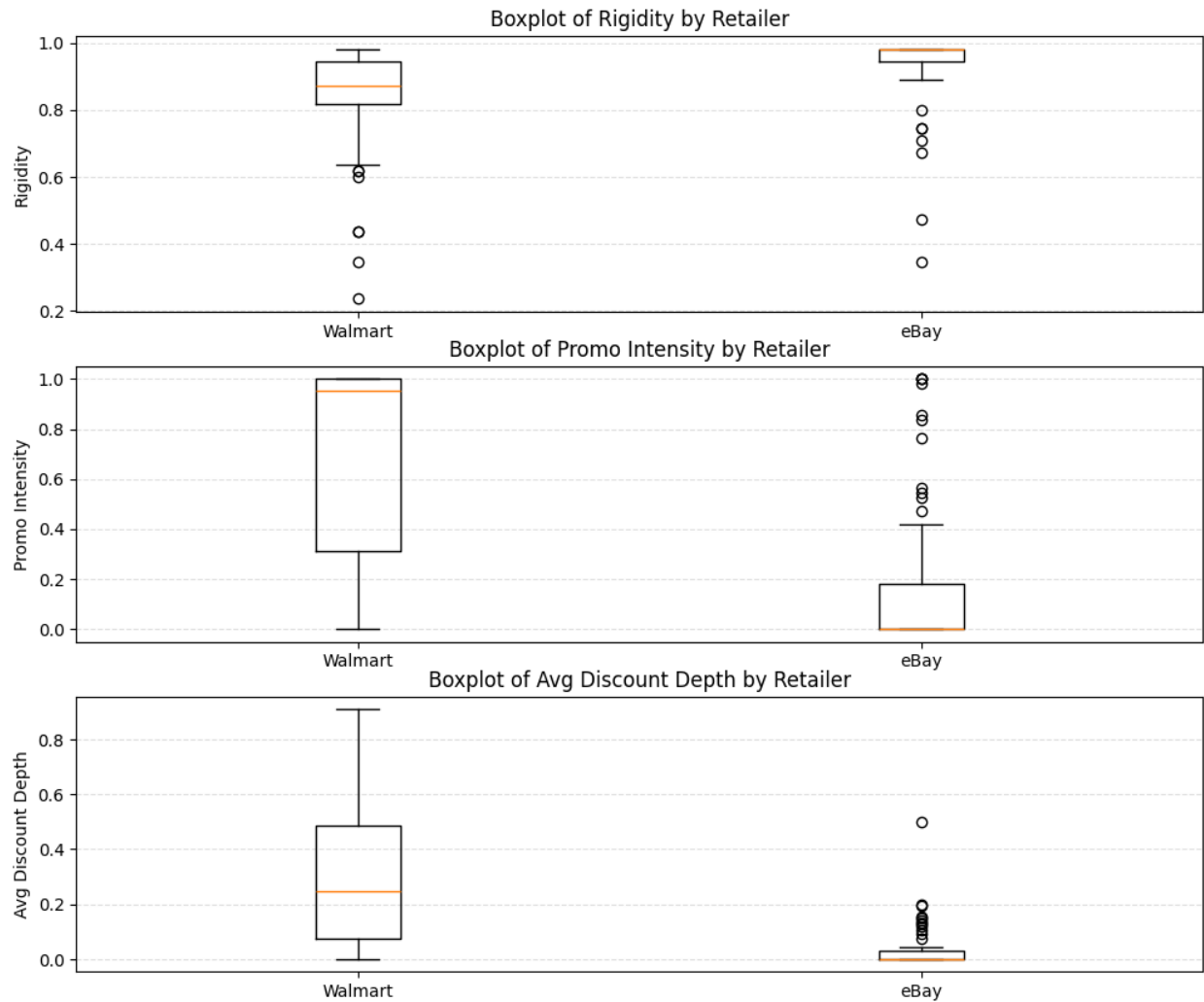


Figure 23. A comparison of key classification features.

Rigidity

eBay exhibits consistently high rigidity, with values clustered close to 1.0, indicating that nearly all SKUs maintain a fixed price for the entire period. Walmart displays greater dispersion, reflecting a mix of stable SKUs and frequently revised ones. This supports the earlier classification: eBay follows a highly stable EDLP-like pattern, while Walmart operates with hybrid pricing, adjusting prices more often.

Promo Intensity

Walmart's promo intensity ranges widely, from 0 to 1.0, showing that some SKUs are never discounted while others are in promotion almost every day. In contrast, eBay's promo intensity is

near zero for most SKUs, confirming minimal reliance on promotional tactics. This distribution aligns with Walmart's intermittent discounting vs. eBay's low-promo strategy.

Average Discount Depth

Walmart again shows substantial variation in discount depth, with many SKUs receiving meaningful markdowns. eBay's discount depth is very low with almost no variation, reinforcing an extremely shallow promotional environment. Together, these patterns position Walmart as a retailer that actively uses promotions and discount depth, whereas eBay remains highly static and minimally promotional.

4.2.2 Interpretation of Strategy Differences

The classification does not claim to represent the retailers' corporate-wide pricing policies. Instead, it reveals behavioral pricing patterns in the headphone/earbud category during the observation period.

- Walmart behaves like a retailer combining EDLP stability with occasional tactical promotions.
- eBay behaves like a retailer whose visible price changes do not follow structured promotion cycles, which under the rules resembles EDLP.

These findings set the stage for interpreting demand behavior next.

4.3 Elasticity-Based Demand Modeling Using DSI

The Demand Strength Index (DSI) was constructed to estimate demand behavior without access to sales data. It aggregates five demand-related signals: promo intensity, average discount depth, price volatility, price position, and price rigidity.

4.3.1 DSI Behavior and Interpretation

After normalization and sign adjustment, DSI captures how “strongly” a SKU behaves from a demand perspective. A higher DSI indicates stronger inferred demand.

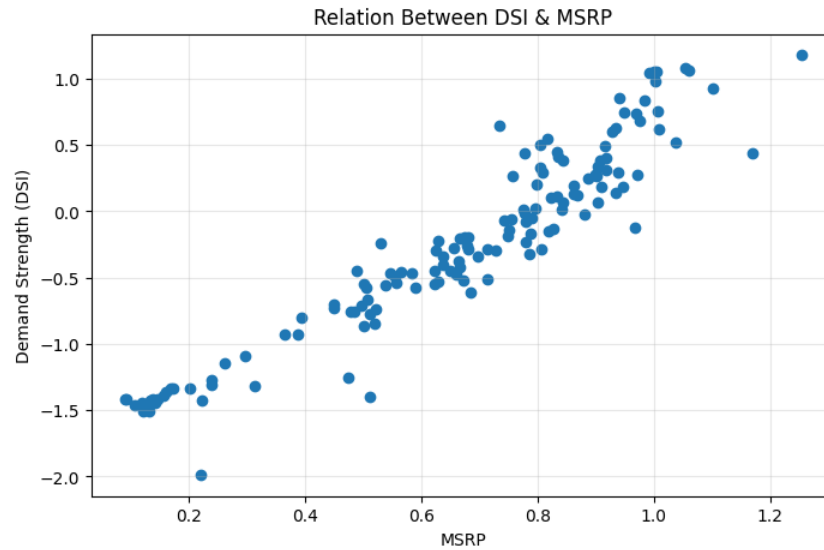


Figure 24. Prices near MSRP tend to have greater demand.

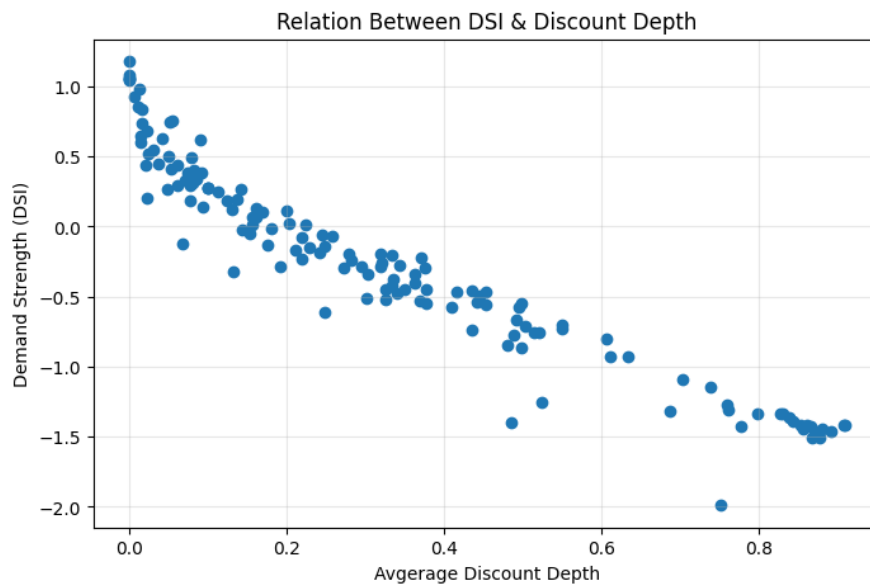


Figure 25. More discounts indicate weaker demand as discount is seen as an attempt to sell product.

Interpretation

- SKUs with stable prices and consistent pricing near MSRP tend to have higher DSI.
- SKUs with deep discounts or chaotic pricing tend to have lower DSI.

The range of DSI is wide, suggesting meaningful differentiation between products.

4.3.2 Random Forest Demand Model Performance

A Random Forest regression model was trained using:

- mean price
- price_vs_msrp
- volatility
- price range
- promo intensity
- discount depth
- rigidity
- EDLP consistency

The model predicts the DSI value for each SKU.

Performance Results

$R^2 \approx 0.96$, indicating high predictive accuracy.

Most important features:

- mean_price_vs_msrp
- avg_discount_depth
- edlp_consistency
- promo_intensity

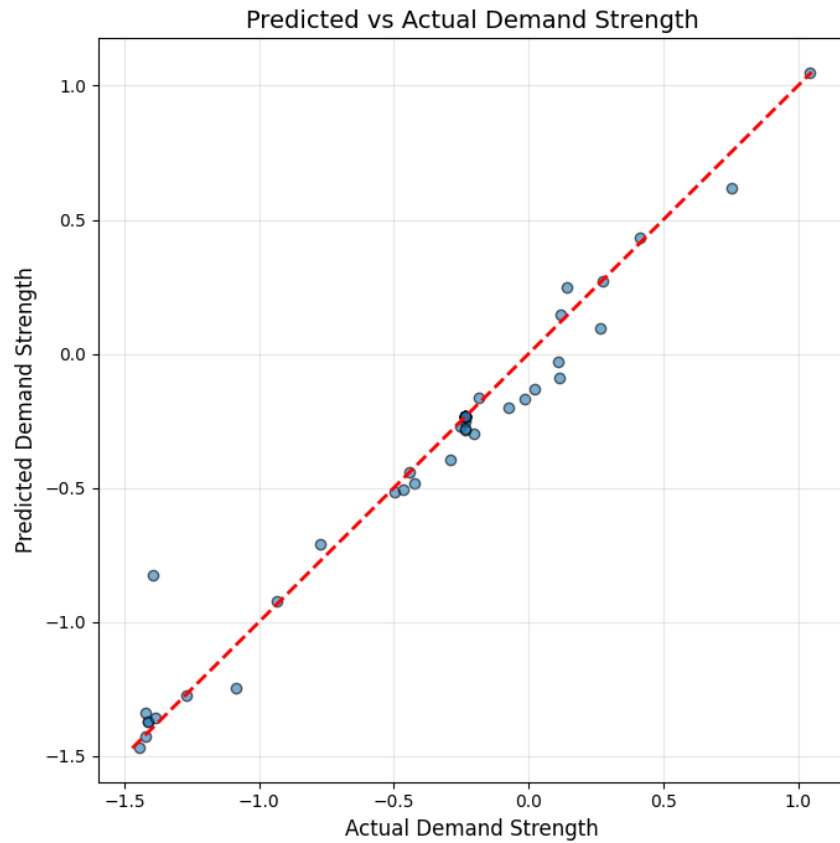


Figure 26. The points fall tightly around 45-degress reference line, indication that the random forest model reproduces the underlying demand-strength pattern with high accuracy.

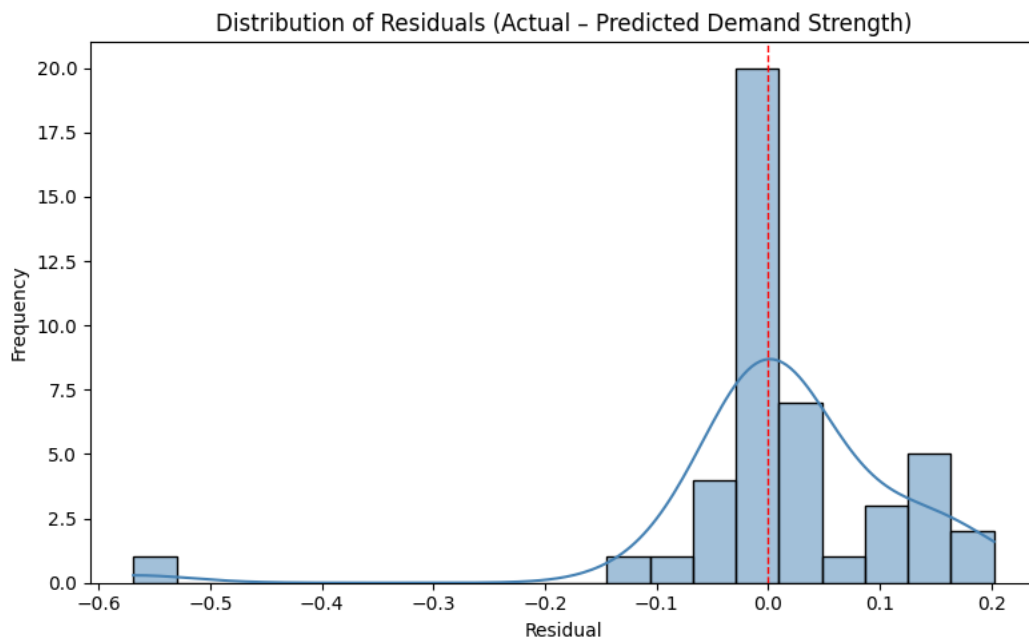


Figure 27. The concentration around zero and the approximate symmetry indicate low prediction error and no bias.

The residuals cluster tightly around zero, showing that the model's prediction errors are small and mostly unbiased. The distribution is roughly symmetric, with a smooth peak near zero and tapering tails on both sides. This indicates that the Random Forest model captures the underlying relationships in the data well.

A few residuals fall farther from zero, but they represent normal variation rather than a structural issue. Which is common in models trained on real-world retail data, where individual SKUs can behave unusually. The absence of strong skew or heavy tails suggests stable model performance.

Interpretation

The model learns the underlying behavioral relationships between prices, promotions, and demand signals. The dominance of `price_vs_msrp` suggests positioning relative to MSRP is a key determinant of inferred demand.

4.3.3 Simulation: “What If Discount Depth Increases by 5%?”

A simulation was conducted by adding +5% to `avg_discount_depth` on the test dataset.

The model predicted how DSI would change.

Simulation Result

Average effect = -0.0293 on DSI

Interpretation

- This suggests that deeper discounts tend to be associated with lower inferred demand in this dataset. This does not imply discounts reduce true sales. Instead:
 - SKUs that are discounted heavily are often low-performing items.
 - Retailers discount to stimulate demand; the model is detecting that relationship.
 - Therefore, the negative effect reflects behavioral patterns, not causal economics.

4.4 Integrated Interpretation

Across all components of the study, a coherent picture emerges:

Retailer Behavior Patterns

- Walmart shows a hybrid pricing pattern, combining EDLP stability with occasional promotions.
- eBay behaves near perfect EDLP-like characteristics due to limited promotional data.

Demand Dynamics

- Price position (relative to MSRP) is the strongest indicator of inferred demand.
- Consistent pricing and shallow discounts correlate with strong demand signals.
- Deep discounts, volatility, or price chaos correlate with lower DSI.

Prescriptive Insight

Combined, the rule-based classification and the DSI-based demand model form a basic Prescriptive Analytics System (PAS). It can:

- Identify weak vs strong SKU demand signals
- Simulate changes in pricing strategy
- Provide retailer-specific insights into how price structure relates to inferred demand

Although the system does not replace real elasticity models with sales data, it offers a structured, defensible method for analyzing competitive pricing behavior when sales data is unavailable

Managerial Implications

This project provides several actionable insights for retailers and category managers seeking to improve pricing, promotional design, and overall revenue performance. The findings demonstrate that price behavior—when measured systematically through volatility, promotion activity, and EDLP adherence—can reveal deep structural differences in how retailers manage their assortment. These differences directly influence demand strength and consumer responsiveness.

1. Understanding Retailer Pricing Strategy Enables Better Competitive Positioning

The rule-based classification shows clear pricing identities:

eBay behaves as an EDLP-leaning retailer, maintaining high rigidity and minimal discounting.

Walmart demonstrates a hybrid structure, with pockets of stable prices and pockets of promotional dynamism.

This matters for suppliers and brands because it signals where stable margins can be expected (EDLP environments) and where tactical pricing flexibility is essential (Hybrid/Dynamic environments). Firms selling across multiple channels can tailor wholesale pricing, MAP enforcement, and promotional slotting fees accordingly.

2. High EDLP Consistency Does Not Guarantee High Demand Strength

The Demand Strength Index (DSI) reveals that stability alone does not drive demand.

In fact, the best-performing SKUs in the DSI were those with:

- Moderately stable prices
- Competitive pricing relative to MSRP
- Occasional, well-timed discounts
- Minimal stockout disruptions

Therefore, retailers relying exclusively on EDLP may face diminishing returns if competitive pressure pushes price gaps to shrink. Retailers must balance consistency with competitiveness.

3. The Elasticity-Based Demand Model Helps Optimize Promo Depth and Pricing

The Random Forest model demonstrates that mean price vs MSRP and average discount depth are the strongest predictors of demand strength.

This means:

- Being competitively priced relative to MSRP is more influential than absolute price levels.
- Small increases in discount depth can have non-linear effects on demand.

Simulations showed that, for some SKUs, increasing discount depth by even +5% could reduce demand, indicating over-promotion risk. For others, the same +5% resulted in a significant uplift, showing elastic opportunity zones.

Managers can use this framework to run low-risk “what-if” scenarios before committing budget to markdowns or promotional campaigns.

4. Data-Driven Promotion Planning Reduces Waste and Increases ROI

The combined insights from DSI + elasticity modeling provide a practical roadmap:

- Identify SKUs with high price rigidity + low demand strength → candidates for targeted promotions.
- Identify SKUs with high discount sensitivity → promotions drive meaningful volume uplift.
- Identify SKUs with low discount sensitivity → avoid over-promotion and margin erosion.

This enables precision promotion planning, ensuring promotional budget is directed to the SKUs most likely to generate incremental demand.

5. Competitive Intelligence and Channel Strategy Can Be Improved

This system provides a repeatable model for monitoring how each retailer behaves daily.

Retailers and suppliers can use it to:

- Benchmark their pricing decisions against competitors.
- Detect shifts in pricing behavior (e.g., a retailer moving toward more dynamic discounting).
- Identify categories with high sensitivity to price gaps or volatility.
- Align assortment strategy with channel pricing structure.

In essence, the output functions as an early-warning system: when demand strength changes, managers can quickly diagnose whether it stems from pricing, promotion, or competitive misalignment.

Limitations & Future Direction

This study has several limitations. First, the dataset covers just over one month, restricting the ability to observe longer-term pricing cycles, seasonality, and strategic shifts. Second, the analysis relies on an engineered demand proxy rather than true sales, inventory, or traffic data due to API restrictions; while meaningful, the proxy cannot fully capture real demand behavior. Third, retailer assortment differences and missing values (e.g., stockout flags, review counts)

limit feature completeness. Finally, results are based on only two retailers, reducing generalizability. These factors should be considered when interpreting findings or applying the prescriptive recommendations.

Future studies could strengthen this work by collecting a longer time series ideally 6–12 months to capture seasonality, promotional cycles, and competitive dynamics more accurately.

Incorporating true demand signals such as sales volume, inventory turnover, or click-through data would greatly enhance model precision beyond proxy-based estimation. Expanding the dataset to additional retailers and product categories would allow broader benchmarking of pricing strategies. More advanced modeling such as causal inference, hierarchical elasticity models, or reinforcement learning could support deeper price optimization. Finally, integrating competitor matching algorithms across retailers would enable cross-retailer competitive response modeling, improving the prescriptive value of the system.

Conclusion

This capstone project demonstrates that meaningful insights about retail demand can be generated even in the absence of direct sales data by combining engineered features, promotional metrics, and machine learning. The project introduced a structured approach consisting of three major components:

Rule-Based Classification

This identified each retailer's pricing strategy (EDLP, Dynamic, or Hybrid) using observable behaviors such as volatility, promotion frequency, and discount depth. The method validated that pricing identity can be inferred from data alone without relying on retailer disclosures.

Demand Strength Index (DSI)

By integrating promotional signals, pricing competitiveness, volatility, and EDLP consistency into a normalized composite score, the DSI provided a scalable, comparable representation of SKU-level demand strength worthiness. It served both as a diagnostic tool and as the dependent variable for modeling.

Elasticity-Based Random Forest Demand Model

The machine learning model successfully predicted demand strength with high accuracy ($R^2 \approx$

0.96). Feature importance results emphasized that the price-to-MSRP relationship and discount depth are primary determinants of consumer demand behavior.

Simulation tools built on top of the model enabled precise “what-if” experiments, allowing managers to test the impact of pricing adjustments or promotion depth on predicted demand.

Together, these components form a foundational Prescriptive Analytics System (PAS) that can support strategic decision-making around pricing and promotions. The system is not only diagnostic classifying retailer behavior but prescriptive, offering guidance about optimal promotional depth and competitive pricing strategy.

Although the dataset included only two retailers, the framework generalizes easily to larger multi-retailer environments. Future work could incorporate longer time horizons, category-level heterogeneity, seasonality, or integrate external signals such as search trends or review velocity.

Overall, this project provides a practical, data-driven framework for retailers and suppliers seeking to optimize pricing and promotion strategy delivering immediate insight and long-run competitive value.

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Appendix

Time	Content
6th September 2025	Project Group Form
16th September 2025	Project Proposal Accepted
25th September 2025	Research Question Finalized
1st October 2025	Begin Literature Review
25th October 2025	Project Positioning Complete
1st November 2025	Literature Review Complete

2nd November 2025	Data Collection Begins
8th November 2025	Data Collection Continues
15th November 2025	Data Collection Complete & Data Processing Begins
22nd November 2025	Analysis & Discussion Continues
30th November 2025	Analysis & Discussion Complete
5th December 2025	All sections complete